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## Introduction

The Rent Manager Web Application Programming Interface (API) offers a powerful and convenient REST-based web services solution for interacting with Rent Manager Online (RMO). The API provides both “read” and “write” access to your database. This means you can develop custom applications that either collect live data from Rent Manager or directly write new data into your system. Here are just a few examples of how the API can be used to extend the capabilities of your property management company:

- Create a website that dynamically updates itself in real-time with changes that occur in Rent Manager.
- Develop a web-based “dashboard” that collects Rent Manager data to display performance metrics on your leasing agents.
- Produce a kiosk-based app where people enter their info and new prospect records are automatically created in Rent Manager.
- Allow another software to upload its information in real-time to Rent Manager.

The RM API makes it possible for you to FULLY access your Rent Manager data. You can use the API to customize the Rent Manager experience of your staff and your clients, and finally, you can integrate external software solutions and custom applications to your Rent Manager database.

## Requirements

In order to develop custom applications with the RM API:

- **You must purchase access to the RM API – contact your Sales Representative for pricing.**
- **You must have an active RMO user account.**
  - This account must have the appropriate user permissions to perform the work of the API application. *For example, in order to write an external kiosk app that writes new prospect records directly into Rent Manager, your user account must have permission to “add prospects.”*
  - If “Locations” are used in Rent Manager, you must have access to the Location to which you want API access. *For example, in order to have that kiosk app write new prospect records into the “SouthBranch” Location, your user account must have access to that Location in Rent Manager.*

## RESTful Api

The Rent Manager Web API adheres to the programming principles of REST (**RE**presentational **State** **T**ransfer). REST-based architecture is focused on **Resources** and **Verbs**.

### Resources

Resources are essentially access points to data entities and RM-specific actions in the Rent Manager database.

| Resource Example | Entity Access                                         |
|------------------|-------------------------------------------------------|
| Tenants          | This resource provides you access to tenant accounts. |

|                  |                                                              |
|------------------|--------------------------------------------------------------|
| Units            | This resource provides you access to unit information.       |
| Vendors          | This resource provides you access to vendor accounts.        |
| RecurringCharges | This resource provides you access to post recurring charges. |

Here are the three types of Resources available in the RM API:

|                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Collection Resource</b> | <p>This is a group (or collection) of records within that Resource. Here are a few examples:</p> <ul style="list-style-type: none"> <li>• All property accounts</li> <li>• All tenant accounts</li> <li>• All vendor accounts</li> </ul> <p>This collection could also be filtered to specific types of records within that Resource:</p> <ul style="list-style-type: none"> <li>• All “active” property accounts</li> <li>• All “past” tenant accounts</li> <li>• All “1099” vendors accounts</li> </ul> |
| <b>Instance Resource</b>   | <p>This is a single record of data within that Resource. Here are a few examples:</p> <ul style="list-style-type: none"> <li>• A single property record</li> <li>• A single rental unit</li> <li>• A single vendor</li> </ul>                                                                                                                                                                                                                                                                             |
| <b>Action Resource</b>     | <p>This is a unique type of Resource that focuses on triggering “work” in Rent Manager. Here are a few examples:</p> <ul style="list-style-type: none"> <li>• Post recurring charges</li> <li>• Post late fees</li> <li>• Merge vendors</li> </ul>                                                                                                                                                                                                                                                        |

## Verbs

**Verbs** determine what should be done with or to the specified Resource.

Here are a few examples:

| Verb          | What does it do to a Collection Resource?                                                      | What does it do to an Instance Resource?                           |
|---------------|------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| <b>Get</b>    | <b>Retrieves</b> that collection of records from the specified Resource.                       | <b>Retrieves</b> that specific record from the specified Resource. |
| <b>Post</b>   | <b>Creates</b> one or more new records of the specified Resource in the Rent Manager database. | <b>Updates</b> that specific record in the Rent Manager database.  |
| <b>Delete</b> | <b>Delete</b> the records from the specified Resource in the Rent Manager database.            | <b>Delete</b> that specific record in the Rent Manager database.   |

You can use GET, POST, and DELETE on *Collection Resources and Instance Resources*.

You can only POST to *Action Resources*.

## Requests

A Request basically translates to: *Do something with or to (Verb) the specified data/action (Resource) in Rent Manager.*

*For more in-depth information on the formal structure of a Request, please refer to the RM API Overview Guide. The “overview” document explains the proper way to write a Request Header and a Request Body (for POST Requests). It further explains how Data Models are collected by GET Requests and how to build Data Models for POST Requests.*

API Requests are written in the form of a URL composed of several components:

Example 1: **GET** <https://corpid.api.rentmanager.com/tenants>

| Sample Component                                | URL Component       | Description                                                                                    |
|-------------------------------------------------|---------------------|------------------------------------------------------------------------------------------------|
| <code>https://corpid.api.rentmanager.com</code> | Protocol + Hostname | This ensures that the request is SSL-encrypted and directs the API to your corporate database. |
| <code>/tenants</code>                           | Resource            | Examine all tenants – a <i>Collection Resource</i> .                                           |

Translation: Retrieve all tenants from the Rent Manager database.

Example 2: **DELETE** `https://corpid.api.rentmanager.com/tenants/35`

| Sample Component                                | URL Component       | Description                                                                                    |
|-------------------------------------------------|---------------------|------------------------------------------------------------------------------------------------|
| <code>https://corpid.api.rentmanager.com</code> | Protocol + Hostname | This ensures that the request is SSL-encrypted and directs the API to your corporate database. |
| <code>/tenants/35</code>                        | Resource            | Examine the tenant with an ID of 35 – an <i>Instance Resource</i> .                            |

Translation: Delete the tenant with an ID of 35 from the Rent Manager database.

Example 3: **POST** `https://corpid.api.rentmanager.com/tenants/35?fields= LastName`

| Sample Component                                | URL Component       | Description                                                                                    |
|-------------------------------------------------|---------------------|------------------------------------------------------------------------------------------------|
| <code>https://corpid.api.rentmanager.com</code> | Protocol + Hostname | This ensures that the request is SSL-encrypted and directs the API to your corporate database. |
| <code>/tenants/35</code>                        | Resource            | Examine the tenant with an ID of 35 – an <i>Instance Resource</i> .                            |
| <code>?fields= LastName</code>                  | Parameter           | Update the LastName field on the specified Resource.                                           |

Translation: Update the tenant with an ID of 35 from the Rent Manager database.

## Sending Requests to the Api

Once you have been authenticated by the API, you can make Requests to the Rent Manager database.

The following example illustrates the retrieval of a single tenant. Download and review the **Rent Manager API Sample Application** for additional information on executing requests and transforming data into usable data models.

|                                |                                                                                                                                                                                                                                                                                      |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>getting a single tenant</b> | <pre>using System.Net.Http; using System.Net.Http.Headers;  static async Task GetObject(string apiToken) {     HttpClient client = new HttpClient();      client.BaseAddress = new Uri("https://corpid.api.rentmanager.com/");     client.DefaultRequestHeaders.Accept.Add(new</pre> |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

```

        MediaTypeWithQualityHeaderValue("application/json"));
        client.DefaultRequestHeaders.Add("X-RM12Api-ApiToken", apiToken);

        HttpResponseMessage response = await client.GetAsync("/tenants/1");
        response.EnsureSuccessStatusCode();
        TenantModel tenant = response.Content.ReadAsAsync<tenantmodel>().Result;

        return tenant;
    }

    public class TenantModel
    {
        public int TenantID { get; set; }
        public string Name { get; set; }
        public string FirstName { get; set; }
        public string LastName { get; set; }
        public bool IsCompany { get; set; }
        public int ColorID { get; set; }
        public string Comment { get; set; }
        public int RentDueDay { get; set; }
        public string LeasePeriod { get; set; }
        public bool DoNotChargeLateFees { get; set; }
        public bool DoNotPrintStatements { get; set; }
        public bool DoNotAcceptChecks { get; set; }
        public bool IsProspect { get; set; }
        public int LeasingAgentID { get; set; }
        public int AccountGroupID { get; set; }
        .
        .
        .
        public int UpdateUserID { get; set; }
    }

```

#### json to model serialization

As shown above, the json returned from the API is automatically serialized to the model you provide. See the Rent Manager API Sample Application for examples using unserialized json data.

## Authenticate with the Api

Prior to any request, you must first “authenticate” with the API to verify that you have permission to access the Rent Manager database. Authentication creates a time-sensitive token which is then included in each future Request to the API. This token verifies that you have continued permission to the Requests that you make to the Rent Manager database. If the token ever becomes invalid or expires, you will be forced to re-authenticate. To authenticate:

|                  |                                                                                                                                        |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| <b>url</b>       | https:// <b>corpid</b> /Authentication/AuthorizeUser<br>(where <b>corpid</b> is the unique identifier assigned to your company by LCS) |
| <b>method</b>    | POST                                                                                                                                   |
| <b>data type</b> | Json                                                                                                                                   |
| <b>sample</b>    | using System.Net.Http;                                                                                                                 |

```

using System.Net.Http.Headers;

string apiToken;

HttpClient client = new HttpClient();
client.BaseAddress = new Uri("https://corpid.api.rentmanager.com/");
client.DefaultRequestHeaders.Accept.Add(new
    MediaTypeWithQualityHeaderValue("application/json"));

// Replace Username, Password, and Location ID with those supplied by LCS.
UserAuthorizationModel uam = new UserAuthorizationModel
{
    Username = "QsUser",
    Password = "123456",
    LocationID = 1
};

HttpResponseMessage response = await
    client.PostAsJsonAsync("/Authentication/AuthorizeUser", uam);
response.EnsureSuccessStatusCode();

apiToken = response.Content.ReadAsStringAsync().Result.Trim("");

// For each additional API request include the apiToken in the Request Header:
client.DefaultRequestHeaders.Add("X-RM12Api-ApiToken", apiToken);

public class UserAuthorizationModel
{
    public string Username { get; set; }
    public string Password { get; set; }
    public int LocationID { get; set; }
}

```

|                   |                                                                                              |
|-------------------|----------------------------------------------------------------------------------------------|
| <b>token</b>      | The apiToken provided by the API is invalidated after 24 hours, or 15 minutes of inactivity. |
| <b>expiration</b> | Once the token expires, re-authentication is required in order to continue API requests.     |

## Rent Manager API Sample Application

The Rent Manager API Sample Application is provided to give you a head start developing your own applications. This is a simple console c# app that illustrates authentication, various resource methods, and even report generation and download. Even if you do not use c#, the techniques should readily transfer to your environment.

To run the Sample Application you will need:

- Access to the internet
- A valid user account authorized to use the API
- Visual Studio 2013 or greater in order to compile and run the samples

You may download the examples from:

<https://corpid.api.rentmanager.com/content/quickstartapp/LCS.UI.API.QuickStart.Examples.zip>  
 (where **corpid** is the unique identifier assigned to your company by LCS)

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